# Newton-Raphson Method for Solving Equations

A numerical method for finding the roots of an arbitrary function using derivatives was first circulated by **Isaac Newton** in 1669. **John Wallis** published Newton's method in 1685, and in 1690 **Joseph Raphson** (1648–1715) published an improved version, which is the standard form used today.

## 1. The Core Idea & Formula

Suppose we are given a function, such as $f(x) = x^3 - 4x^2 - x + 2$. A graph of this function reveals three roots located near −1, 1, and 4. Because exact algebraic solutions can be difficult or impossible to find, we use an iterative procedure that progressively approximates the true root.

If $x_k$ is our current estimate, and neither $f(x_k)$ nor $f'(x_k)$ is zero, the next improved approximation is given by the recurrence relation:

$$x_{k+1} = x_k - \left[ f(x_k) / f'(x_k) \right]$$

If we seek the root close to −1 for the cubic function above, the successive iterations (with 10-digit accuracy) demonstrate rapid convergence:

- $x_2 = -0.8$
- $x_3 = -0.7628415301$
- $x_4 = -0.7615586962$
- $x_5 = -0.7615571818$ | $x_6 = -0.7615571818$

This provides an exceptionally accurate approximation to 10 decimal places. Note that starting closer to alternate roots yields convergence for those specific values, whereas poor initial estimates require additional steps.

## 2. Worked Examples

**Example 1: Approximating Square Roots**

Use the Newton-Raphson method with a starting point of $x_0 = 3$ to find a solution within $10^{-8}$ of $\sqrt{10}$. The number $\sqrt{10}$ is the positive root of $f(x) = x^2 - 10 = 0$. With $f'(x) = 2x$, the recurrence is:

$$x_{n+1} = x_n - [\ (x_n^2 - 10) / (2x_n)\ ] = \tfrac{1}{2}\ (\ x_n + 10 / x_n\ )$$

Starting with $x_0 = 3$: $x_1 = 3.166666667$, $x_2 = 3.162280702$, $x_3 = 3.16227766$, and $x_4 = 3.16227766$. Since $x_3 = x_4$, convergence is achieved to 8 decimal places.

**Example 2: Solving Polynomial Equations**

Newton's equation $y^3 - 2y - 5 = 0$ has a root near $y = 2$. Starting with $y_0 = 2$, let $f(y) = y^3 - 2y - 5$ and $f'(y) = 3y^2 - 2$. The recurrence formula is:

$$y_{n+1} = [\ 2y_n^3 + 5\ ] / [\ 3y_n^2 - 2\ ]$$

Starting with $y_0 = 2$, we compute $y_1 = 2.1$, $y_2 = 2.094568121$, and $y_3 = 2.094551482$.

**Example 3: Solving Transcendental Equations**

Find all solutions of $e^{2x} = x + 6$ correct to 4 decimal places. Let $f(x) = e^{2x} - x - 6$, so $f'(x) = 2e^{2x} - 1$. The recurrence relation is:

$$x_{n+1} = [\ 2x_n - 1\ ]e^{2x_n} + 6 / [\ 2e^{2x_n} - 1\ ]$$

Choosing an initial estimate $x_0 = 0.97$ yields $x_1 = 0.970870836$ and $x_2 = 0.97087002$. Because $x_1$ and $x_2$ agree to 5 decimal places, the solution correct to 4 decimal places is **0.9709**.

## Exercise

Let $f(x) = x^2 - a$. Show that applying the Newton-Raphson method leads to the general recurrence relation: $x_{n+1} = \tfrac{1}{2}\ (\ x_n + a / x_n\ )$.